Informatica® Cloud Application Integration
April 2025

Google Vertex AI Connector Guide

Informatica Cloud Application Integration Google Vertex AI Connector Guide
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Preface

Read the *Google Vertex AI Connector Guide* to learn how to set up and use Google Vertex AI connections. This guide assumes that you have an understanding of the Google Vertex AI concepts.

CHAPTER 1

Introduction to Google Vertex AI Connector

Use Google Vertex AI Connector to easily integrate Vertex AI with various data sources and tools, increasing the functionality and reach of your machine learning workflows.

You can use Google Vertex AI Connector to achieve use cases, such as data ingestion, feature engineering, model training and validation, and business intelligence integration.

Google Vertex AI Connector provides the following key features:

- Streamlines the flow of data between Vertex AI and other Google Cloud services, such as BigQuery, Google Cloud Storage, and Dataproc, allowing for seamless data management and processing.
- Enables Vertex AI to connect with external data sources, enhancing the ability to build and train models on diverse datasets.
- Provides a smooth data integration experience, reducing the manual workload of moving and preparing data for machine learning tasks.
- Supports live data streams, ensuring that models can be trained and updated with the latest information.
- Facilitates automated workflows by integrating with other tools, helping to streamline model development, deployment, and monitoring processes.
- Provides a unified interface to access various data sources and manage machine learning (ML) workflows.
- Ensures secure data handling and compliance with industry standards when working with sensitive information.

Google Vertex AI overview

Google Vertex AI is a managed machine learning platform provided by Google Cloud that enables users to build, deploy, and scale machine learning models. It is designed to streamline the development and deployment of ML models. Vertex AI combines Google Cloud's powerful services into a unified environment.

Google Vertex AI provides the following key features:

- Provides a single interface to manage all stages of the ML workflow, including data preprocessing, model training, verification, deployment, and monitoring.
- Includes access to pre-trained APIs for common tasks, such as image analysis and text processing.
- Supports custom model development using frameworks, such as TensorFlow, PyTorch, and Scikit-learn. It simplifies the training process with a managed infrastructure.

- Offers tools for managing model operations (MLOps) to automate and streamline the deployment, scaling, and maintenance of models.
- Vertex AI's AutoML feature automates the model training process, enabling users with minimal ML expertise to build models effectively.
- Provides scalable compute resources, allowing models to be trained and deployed on an infrastructure that automatically scales according to demand.
- Simplifies versioning, deployment, and monitoring of models, ensuring that they perform as expected in production environments.
- Offers tools to understand model predictions and eliminate possible issues.
- Integrates with other Google Cloud services and supports collaboration.

CHAPTER 2

Google Vertex AI connections

Use a Google Vertex AI connection to integrate with various data sources and services, and build comprehensive and flexible machine learning workflows.

You can perform various actions, such as generating content, function calling, grounding, text embeddings, and multimodal embeddings using the access token and model ID.

After you create a Google Vertex AI connection, you can validate, test, and save the connection.

You can then publish the Google Vertex AI connection and click the **Metadata** tab to view the generated process objects for the connection.

Basic connection properties

The following table describes the basic properties that you can configure on the **Properties** tab of the connection creation page:

Property	Description
Name	Required. Unique name for the Google Vertex AI connection that identifies it in the Process Designer. The name must start with an alphabet and can contain only alphabets, numbers, and hyphens (-).
Location	The location of the project or folder where you want to save the connection. Click **Browse** to select a location. If the **Explore** page is currently active and a project or folder is selected, the default location for the connection is the selected project or folder. Otherwise, the default location is the location of the most recently saved asset.
Description	Optional. Description of the connection.
Type	Required. The type of connection you want to use for the connector or service connector. Select **Google Vertex AI**.
Runtime Environment	Required. The runtime environment for the connection. You can run the connection on the Cloud Server, a Secure Agent group, or a Secure Agent machine.
Connection Test	Not supported for Google Vertex AI Connector.
OData-Enabled	Not supported for Google Vertex AI Connector.

Google Vertex AI connection properties

To use Google Vertex AI, you need the project ID and private key to access the service account.

The following table defines the Google Vertex AI connection properties that you must configure in the **Connection Properties** section:

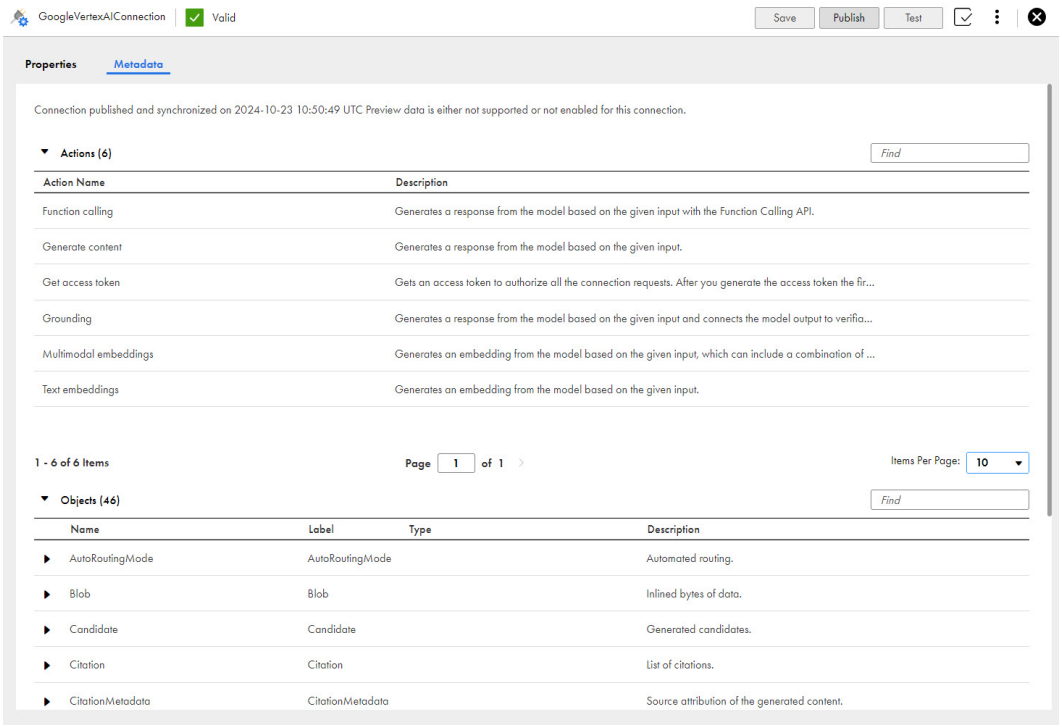
Property	Description
Project_ID	The project ID to generate a valid access token.
Location	The region to process the request.
Private_Key_ID	The private key ID associated with the service account.
Client_Email	The email address associated with the service account.
Private_Key	The Google Vertex AI private key associated with the service account. Enter the PKCS1 certificate as a Base64-encoded string in the following format: `-----BEGIN PRIVATE KEY----- ..... n-----END PRIVATE KEY-----`

Google Vertex AI connection metadata

After you create a Google Vertex AI connection, you can validate and save the connection. You can then publish the Google Vertex AI connection and click the **Metadata** tab to view the generated process objects for the connection.

When you publish a Google Vertex AI connection in Application Integration, the **Actions** and **Objects** appear on the **Metadata** tab.

The following image shows the **Metadata** tab for a published Google Vertex AI connection:



Publishing Google Vertex AI connections

When you publish a Google Vertex AI connection in Application Integration, the **Actions** and **Objects** appear on the **Metadata** tab.

Consider the following information when you use specific actions and objects:

- The **Generate content** action generates a response from the model based on the given input. It creates a *requestBody* that contains fields with different types, such as string or array.
  You must insert the access_token field from the **Get access token** action and the model_ID field to call the model, and create a payload. For example, the model_ID value can be `gemini-1.5-flash-001` or `gemini-1.5-pro-001`.

  The following snippet is a sample to build a `requestBody` object:

```
<root>
    <contents>
<role>user</role>
     <parts>
         <text>Describe what you hear in this audio</text>
     </parts>
     <parts>
         <inlineData>
             <mimeType>audio/aac</mimeType>
             <data>{BASE64ENCODED}</data>
         </inlineData>
     </parts>
</contents>
<generationConfig>
     <temperature>0.5</temperature>
     <maxOutputTokens>200</maxOutputTokens>
     <topP>0.8</topP>
     <topK>10</topK>
```

```
</generationConfig>
</root>
```

If fields in the `requestBody` need to be in different formats, you must add additional namespaces as shown in the following sample code:

```
<root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
    <field_array m:isArray="true">
        <property_1> ...</property_1>
    </field_array>
    <field_boolean m:type="xs:boolean">true</field_boolean>
    <field_double m:type="xs:double">20.2</field_double>
    <field_integer m:type="xs:int">10</field_integer>
</root>
```

- The **Function calling** action generates a response from the model based on the given input with the Function Calling API.
  You must insert the access_token field from the **Get access token** action and the model_ID field to call the model, and create a payload. For example, the model_ID value can be `gemini-1.5-flash-001` or `gemini-1.5-pro-001`.

  The following snippet is a sample to build a `requestBody` object:

```
<root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
    <contents>
        <role>user</role>
        <parts>
            <text>What is the weather in Boston, MA?</text>
        </parts>
    </contents>
    <tools>
        <functionDeclarations>
            <name>get_current_weather</name>
            <description>Get the current weather in a given location</description>
            <parameters>
                <type>object</type>
                <properties>
                    <location>
                        <type>string</type>
                        <description>The city and state, e.g. San Francisco, CA or a
zip code e.g. 95616</description>
                    </location>
                </properties>
                <required m:isArray="true">location</required>
            </parameters>
        </functionDeclarations>
    </tools>
</root>
```

  If fields in the `requestBody` need to be in different formats, you must add additional namespaces as shown in the following sample code:

```
<root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
    <field_array m:isArray="true">
        <property_1> ...</property_1>
    </field_array>
    <field_boolean m:type="xs:boolean">true</field_boolean>
    <field_double m:type="xs:double">20.2</field_double>
    <field_integer m:type="xs:int">10</field_integer>
</root>
```

- The **Grounding** action generates a response from the model based on the given input and connects the model output to verifiable sources of information.
  You must insert the access_token field from the **Get access token** action and the model_ID field to call the model, and create a payload. For example, the model_ID value can be `gemini-1.5-flash-001` or `gemini-1.5-pro-001`.

The following snippet is a sample to build a `requestBody` object:

```
<root>
    <contents>
        <role>user</role>
        <parts>
            <text>Can you provide the latest information on climate change's impact
on polar bears?</text>
        </parts>
    </contents>
    <tools>
        <googleSearchRetrieval />
    </tools>
</root>
```

If fields in the `requestBody` need to be in different formats, you must add additional namespaces as shown in the following sample code:

```
<root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
    <field_array m:isArray="true">
        <property_1> ...</property_1>
    </field_array>
    <field_boolean m:type="xs:boolean">true</field_boolean>
    <field_double m:type="xs:double">20.2</field_double>
    <field_integer m:type="xs:int">10</field_integer>
</root>
```

- The **Text embeddings** action generates an embedding from the model based on the given input.
  You must insert the access_token field from the **Get access token** action and the model_ID field to call the model, and create a payload. For example, the model_ID value can be `textembedding-gecko-multilingual@001` or `text-multilingual-embedding-002`.

  The following snippet is a sample to build a `requestBody` object:

  ```
  <root>
      <instances>
          <task_type>RETRIEVAL_DOCUMENT</task_type>
          <title>document title</title>
          <content>I would like embeddings for this text!</content>
      </instances>
  </root>
  ```

  If fields in the `requestBody` need to be in different formats, you must add additional namespaces as shown in the following sample code:

  ```
  <root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
      <field_array m:isArray="true">
          <property_1> ...</property_1>
      </field_array>
      <field_boolean m:type="xs:boolean">true</field_boolean>
      <field_double m:type="xs:double">20.2</field_double>
      <field_integer m:type="xs:int">10</field_integer>
  </root>
  ```

- The **Multimodal embeddings** action generates an embedding from the model based on the given input, which can include a combination of image, text, and video data.
  You must insert the access_token field from the **Get access token** action and the model_ID field to call the model, and create a payload. For example, the model_ID value can be `multimodalembedding@001`.

  The following snippet is a sample to build a `requestBody` object:

  ```
  <root>
      <instances>
          <text>Describe what you see</text>
          <video>
              <bytesBase64Encoded>{BASE64ENCODED}</bytesBase64Encoded>
              <videoSegmentConfig>
                  <endOffsetSec>1</endOffsetSec>
              </videoSegmentConfig>
          </video>
          <parameters>
  ```

```
            <dimension>1536</dimension>
        </parameters>
    </instances>
</root>
```

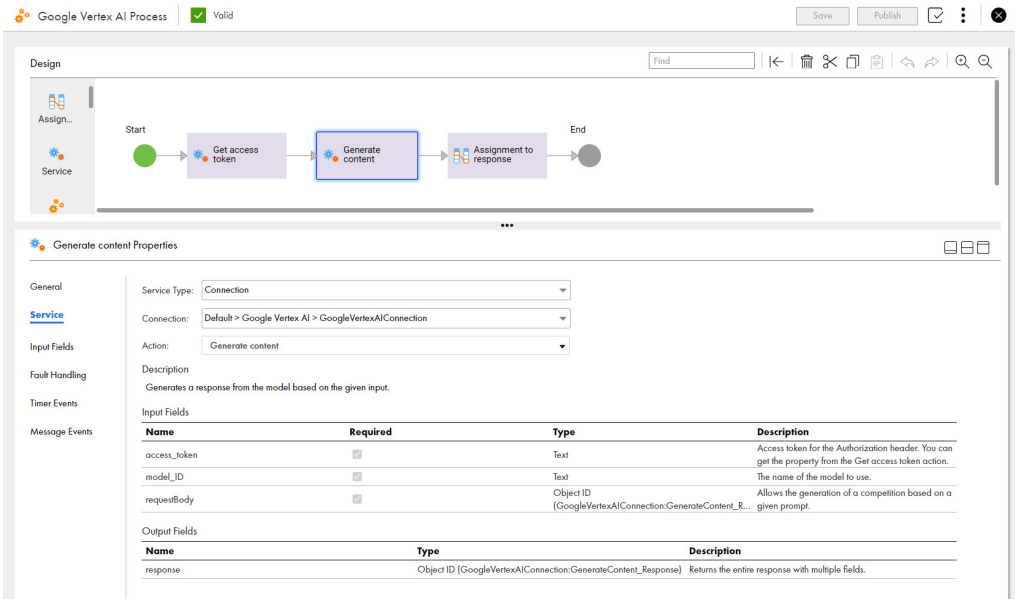
If fields in the `requestBody` need to be in different formats, you must add additional namespaces as shown in the following sample code:

```
<root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
    <field_array m:isArray="true">
        <property_1> ...</property_1>
    </field_array>
    <field_boolean m:type="xs:boolean">true</field_boolean>
    <field_double m:type="xs:double">20.2</field_double>
    <field_integer m:type="xs:int">10</field_integer>
</root>
```

Using a Google Vertex AI connection in a process

After you create a Google Vertex AI connection, you can use it in a Service step in a process.

1. Add a Service step to the process.
2. Click the **Service** tab.
3. From the **Service Type** list, select **Connection**.
4. From the **Connection** list, browse and select the Google Vertex AI connection that you created.
5. From the **Action** list, select the action that you want to perform. For example, select the **Generate content** action to generate a response from the model based on the given input as shown in the following image:



In this case, Application Integration populates the **access_token**, **model_ID**, and **requestBody** objects under the input fields and **response** under the output fields.

6. Configure further steps as needed.
7. Validate, save, and publish the process.